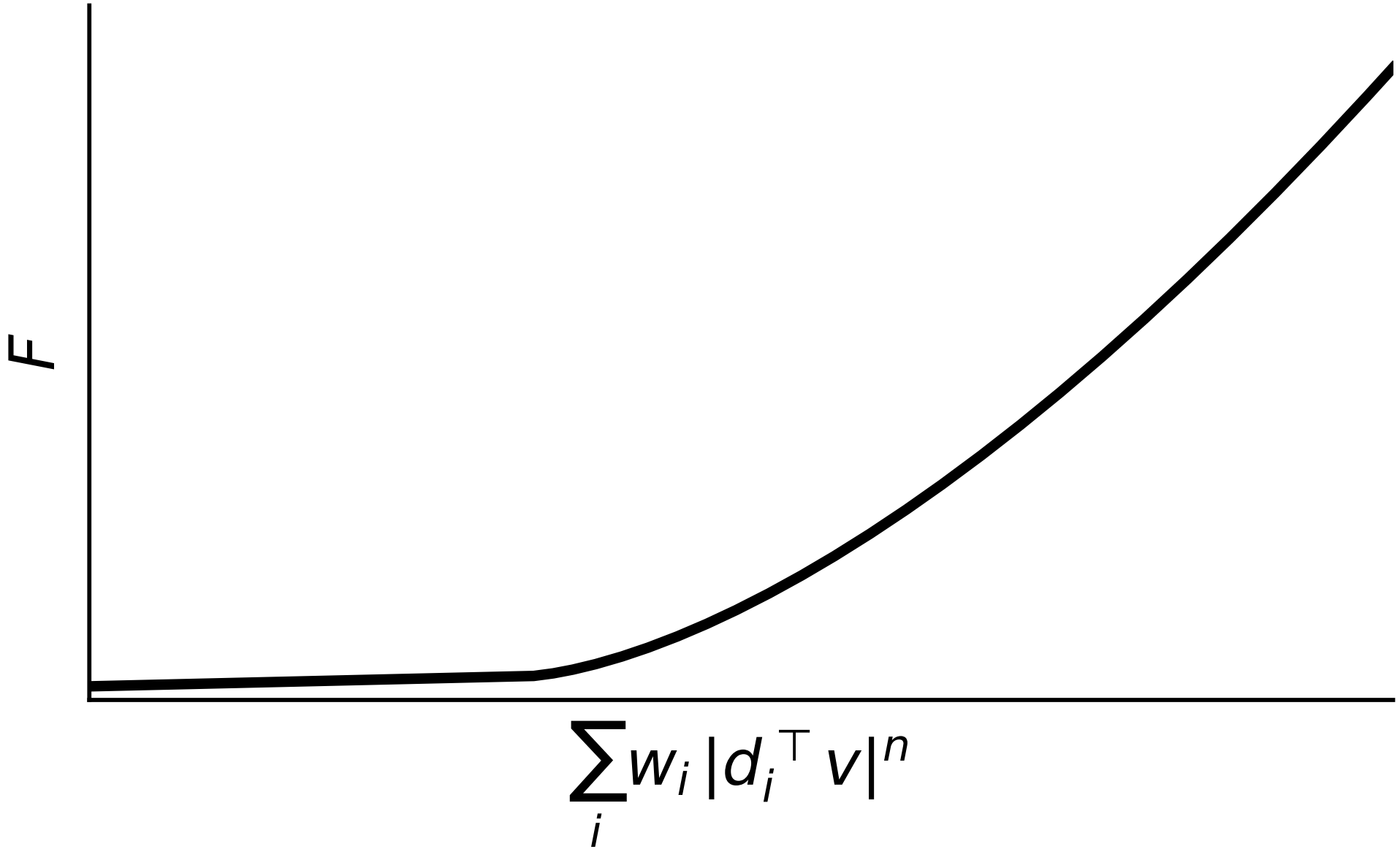
F
$\sum_i w_i \, |d_i{}^\top v|^n$